WEEK 08 QUESTION BANK

1. Develop a Python program using the **Logistic Regression algorithm** to classify a tumour as malignant or benign based on medical measurements using the Breast Cancer Wisconsin dataset.
2. Develop a Python program using the **Logistic Regression algorithm** to predict whether a person is diabetic or non-diabetic based on medical attributes using the Pima Indians Diabetes Dataset.
3. Build a **Support Vector Machine (SVM) classifier** to predict fish species based on physical measurements using a Fish dataset obtained from Kaggle.
4. Build **Random Forest-based model** in python for Breast Cancer Wisconsin (diagnostic) dataset from sci-kit learn Or dataset from UCI , Kaggle.
5. Explain the sigmoid function in Logistic Regression with its mathematical formula, graph, and suitable examples of probability values.
6. Differentiate between Linear Regression and Logistic Regression with suitable examples.
7. Explain the overview of Logistic Regression and describe how it is used for classification problems.
8. Explain any five assumptions of Logistic Regression.
9. Describe any five real-world applications of Logistic Regression.
10. Explain the concept of Support Vector Machine (SVM) and describe its importance in machine learning.
11. Explain the working procedure of Support Vector Machine (SVM) with a neat diagram, describing the hyperplane, support vectors, and maximum margin.
12. Differentiate between Linear SVM and Non-Linear SVM based on data separability and classification boundaries
13. Differentiate between Stacking and Blending with suitable examples.
14. Differentiate between Max Voting, Averaging, and Weighted Average techniques with suitable examples.
15. Explain how Bagging can be applied to improve the performance of a machine learning model using different training samples.
16. Explain how Boosting improves the performance of weak learners with a suitable example.
17. Describe the working procedure of the Random Forest algorithm, explaining how multiple decision trees contribute to the final prediction.
18. A bank wants to develop a machine learning system to predict loan default risk. Explain how Random Forest can be applied to this problem and discuss its advantages and limitations